

Practice Questions for 2024-10-22-Regression

Generated by Scribe.AI

December 31, 2024

1 Questions

1. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
 - a. 5
 - b. 4
 - c. 5.5
 - d. 6
 - e. 7

2. In the context of L^1 -regression (least absolute deviations), what is the significance of formulating the problem as a linear program, and what advantages does this offer compared to other solution methods?
 - a. The linear program formulation simplifies the problem, making it easier to solve manually.
 - b. The linear program formulation allows for the use of efficient and robust linear programming algorithms to find the optimal solution, offering scalability and guaranteed optimality.
 - c. The linear program formulation is only advantageous for small datasets; for larger datasets, other methods are more efficient.
 - d. The linear program formulation provides an approximate solution, which is sufficient for most practical applications.
 - e. The linear program formulation is computationally more expensive than other methods, but it provides a more accurate solution.

3. How does the L^∞ -regression (minimax) problem differ conceptually from L^1 and L^2 regression, and how does this difference manifest in its linear programming formulation?
 - a. The L^∞ regression minimizes the sum of squared errors, while L^1 and L^2 minimize the sum of absolute errors.
 - b. The L^∞ regression minimizes the maximum absolute error, unlike L^1 and L^2 which focus on the sum of errors.
 - c. The L^∞ regression is not suitable for linear programming formulation.
 - d. The L^∞ regression is equivalent to L^1 regression when the data is normally distributed.
 - e. The L^∞ regression is computationally more expensive than L^1 and L^2 regression, regardless of the formulation.

4. In the context of binary classification using Support Vector Machines (SVMs), explain the geometric interpretation of maximizing the margin and how this relates to the linear programming formulation presented in the slides.
 - a. Maximizing the margin means finding the hyperplane that minimizes the distance to the closest data points of both classes.
 - b. Maximizing the margin is equivalent to minimizing the sum of distances from all data points to the hyperplane.
 - c. Maximizing the margin means finding the hyperplane that maximizes the distance to the furthest data point from either class.
 - d. Maximizing the margin is a heuristic approach and does not have a direct relationship to the linear programming formulation.
 - e. Maximizing the margin means finding the hyperplane that maximizes the minimum distance to the closest data points of both classes.
5. Explain the differences between L^1 , L^2 , and L^∞ regression, focusing on their objective functions and the resulting solutions. How does the choice of error metric affect the robustness of the model to outliers?
 - a. L^1 regression minimizes the sum of squared errors, leading to the mean; L^2 minimizes the sum of absolute errors, leading to the median; L^∞ minimizes the maximum absolute error, leading to the mid-range. L^1 is most robust to outliers.
 - b. L^1 regression minimizes the sum of absolute errors, leading to the median; L^2 minimizes the sum of squared errors, leading to the mean; L^∞ minimizes the maximum absolute error, leading to the mid-range. L^1 is most robust to outliers.
 - c. L^1 regression minimizes the maximum absolute error, leading to the mid-range; L^2 minimizes the sum of squared errors, leading to the mean; L^∞ minimizes the sum of absolute errors, leading to the median. L^∞ is most robust to outliers.
 - d. L^1 regression minimizes the sum of absolute errors, leading to the median; L^2 minimizes the sum of squared errors, leading to the mean; L^∞ minimizes the maximum absolute error, leading to the mid-range. L^∞ is most robust to outliers.
 - e. L^1 regression minimizes the sum of squared errors, leading to the mean; L^2 minimizes the sum of absolute errors, leading to the median; L^∞ minimizes the maximum absolute error, leading to the mid-range. L^∞ is most robust to outliers.
6. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
 - a. 1
 - b. 3
 - c. 5
 - d. 7
 - e. 9
7. In the context of L^1 -regression (least absolute deviations), what is the significance of formulating the problem as a linear program, and what advantages does this offer compared to other solution methods?

- a. It simplifies the problem but is less efficient than other methods.
 - b. It allows for the use of efficient linear programming algorithms to find the optimal solution.
 - c. It is only applicable to small datasets.
 - d. It provides a less accurate solution compared to other methods.
 - e. It is computationally more complex than other methods.
8. How does the L^∞ -regression (minimax) problem differ conceptually from L^1 and L^2 regression, and how does this difference manifest in its linear programming formulation?
- a. It minimizes the sum of squared errors, unlike L^1 and L^2 which minimize the maximum error.
 - b. It minimizes the maximum absolute error, unlike L^1 and L^2 which minimize the sum of absolute or squared errors.
 - c. It minimizes the median absolute error, unlike L^1 and L^2 which minimize the mean absolute or squared errors.
 - d. There is no conceptual difference; only the notation changes in the linear programming formulation.
 - e. It minimizes the average error, unlike L^1 and L^2 which minimize the maximum error.
9. In the context of binary classification using Support Vector Machines (SVMs), explain the geometric interpretation of maximizing the margin and how this relates to the linear programming formulation presented in the slides.
- a. Maximizing the margin means minimizing the distance between the separating hyperplane and the closest data points of each class.
 - b. Maximizing the margin means maximizing the distance between the separating hyperplane and all data points.
 - c. Maximizing the margin means minimizing the number of misclassified points.
 - d. Maximizing the margin has no geometric interpretation; it's purely an algebraic concept.
 - e. Maximizing the margin means finding the hyperplane that passes through the origin.
10. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
- a. 1
 - b. 3
 - c. 5
 - d. 7
 - e. 9

2 Answers

1. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
 - a. The L^1 -error minimization leads to the median. The median of the data set $\{1, 3, 5, 7, 9\}$ is 5.
 - b. The mean of the data set is 5, but the L^1 error minimization yields the median, not the mean.
 - c. This is the average of the minimum and maximum values (mid-range), which corresponds to L^∞ error minimization.
 - d. This is neither the mean nor the median of the data set.
 - e. This is neither the mean nor the median of the data set.

2. In the context of L^1 -regression (least absolute deviations), what is the significance of formulating the problem as a linear program, and what advantages does this offer compared to other solution methods?
 - a. While the linear program formulation simplifies the problem conceptually, it doesn't necessarily make it easier to solve manually. Linear programming solvers are typically used.
 - b. The linear program formulation of L^1 -regression leverages the power of established linear programming algorithms (like the simplex method or interior-point methods). These algorithms are highly optimized and can efficiently solve large-scale problems, guaranteeing an optimal solution. Other methods might be faster for very small datasets, but they lack the scalability and optimality guarantees of linear programming.
 - c. The opposite is true. Linear programming solvers are designed to handle large datasets efficiently.
 - d. Linear programming provides the exact optimal solution, not an approximation.
 - e. Linear programming solvers are generally efficient and can handle large problems effectively. While the initial formulation might seem more complex, the computational cost is often lower than other methods for larger datasets due to the efficiency of the solvers.

3. How does the L^∞ -regression (minimax) problem differ conceptually from L^1 and L^2 regression, and how does this difference manifest in its linear programming formulation?
 - a. This statement incorrectly swaps the characteristics of L^2 and L^1 regression.
 - b. The core difference is that L^∞ regression aims to minimize the largest absolute error, a minimax approach. L^1 and L^2 minimize the sum of absolute or squared errors, respectively. In the linear program formulation, this difference is reflected in the objective function: L^∞ minimizes a single slack variable representing the maximum error, while L^1 and L^2 minimize the sum of slack variables or a quadratic function of errors.
 - c. The L^∞ regression can be effectively formulated as a linear program, as shown in the slides.
 - d. There's no such equivalence between L^1 and L^∞ regression, regardless of data distribution.
 - e. The computational cost depends on the solver and the size of the problem. The linear programming formulations of all three allow for efficient solution using specialized solvers.

4. In the context of binary classification using Support Vector Machines (SVMs), explain the geometric interpretation of maximizing the margin and how this relates to the linear programming formulation presented in the slides.

- a. This describes minimizing the margin, the opposite of the SVM goal.
 - b. This describes a different optimization problem, not the margin maximization of SVMs.
 - c. This describes a different optimization problem, not the margin maximization of SVMs.
 - d. Maximizing the margin is the central concept of SVMs, and the linear program formulation directly implements this goal.
 - e. The geometric interpretation of maximizing the margin is to find the hyperplane that creates the largest possible separation between the two classes. This is achieved by maximizing the minimum distance from the hyperplane to the closest data points of each class. The linear program formulation directly reflects this by maximizing a slack variable (δ) representing this minimum distance, subject to constraints ensuring that all points are at least a distance δ from the hyperplane.
5. Explain the differences between L^1 , L^2 , and L^∞ regression, focusing on their objective functions and the resulting solutions. How does the choice of error metric affect the robustness of the model to outliers?
- a. Answer A incorrectly assigns the objective functions and solutions. While it correctly states that L^1 is most robust, the other assignments are wrong.
 - b. Answer B correctly identifies the objective functions and resulting solutions for each regression type. L^1 regression (minimizing the sum of absolute errors) is more robust to outliers because it is less sensitive to large errors than L^2 regression (minimizing the sum of squared errors). L^∞ regression is the least robust to outliers because it focuses on the single largest error.
 - c. Answer C incorrectly assigns the objective functions and solutions and incorrectly identifies L^∞ as the most robust.
 - d. Answer D correctly identifies the objective functions and solutions but incorrectly identifies L^∞ as the most robust.
 - e. Answer E incorrectly assigns the objective functions and solutions and incorrectly identifies L^∞ as the most robust.
6. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
- a. 1 is the minimum value in the dataset, not the median.
 - b. 3 is not the median of the dataset.
 - c. The L^1 error minimization leads to the median. The median of the dataset $\{1, 3, 5, 7, 9\}$ is 5.
 - d. 7 is not the median of the dataset.
 - e. 9 is the maximum value in the dataset, not the median.
7. In the context of L^1 -regression (least absolute deviations), what is the significance of formulating the problem as a linear program, and what advantages does this offer compared to other solution methods?
- a. Linear programming provides efficient solutions; it doesn't make the problem less efficient.
 - b. Formulating the L^1 regression problem as a linear program allows us to leverage the highly developed and efficient algorithms available for solving linear programs. This is a significant advantage over other methods that might require iterative approaches or specialized algorithms.

- c. Linear programming can handle large datasets, although computational time increases with size.
 - d. The solution is as accurate as the data allows; the method itself doesn't compromise accuracy.
 - e. Linear programming algorithms are generally efficient and well-optimized.
8. How does the L^∞ -regression (minimax) problem differ conceptually from L^1 and L^2 regression, and how does this difference manifest in its linear programming formulation?
- a. The L^∞ method minimizes the maximum error, not the sum of squared errors.
 - b. The L^∞ regression minimizes the maximum absolute error, focusing on reducing the largest error rather than the sum of errors like L^1 and L^2 . In its linear programming formulation, this is reflected by the objective function minimizing a single slack variable representing the maximum error, subject to constraints ensuring that all errors are less than or equal to this maximum.
 - c. The L^∞ method minimizes the maximum error, not the median error.
 - d. There is a significant conceptual difference in the objective function and how it's handled in the linear program.
 - e. The L^∞ method minimizes the maximum error, not the average error.
9. In the context of binary classification using Support Vector Machines (SVMs), explain the geometric interpretation of maximizing the margin and how this relates to the linear programming formulation presented in the slides.
- a. Geometrically, maximizing the margin means finding the separating hyperplane that is maximally distant from the closest data points of each class. This distance is the margin. The linear programming formulation reflects this by having an objective function that maximizes the margin (a variable representing the minimum distance) subject to constraints that ensure all data points are correctly classified and maintain a minimum distance from the hyperplane.
 - b. Maximizing the margin focuses on the minimum distance, not the distance to all points.
 - c. Minimizing misclassified points is a different objective; SVMs aim to maximize the margin for better generalization.
 - d. Maximizing the margin has a clear geometric interpretation related to the distance between the hyperplane and the data points.
 - e. The hyperplane's position is not constrained to pass through the origin in standard SVMs.
10. Given the data points $\{x_1, x_2, x_3, x_4, x_5\} = \{1, 3, 5, 7, 9\}$, what is the solution obtained by minimizing the L^1 -error (least absolute deviations) as shown in Slide 1?
- a. 1 is the minimum value in the dataset, not the median.
 - b. 3 is not the median of the dataset.
 - c. The L^1 error minimization leads to the median of the data set. The median of $\{1, 3, 5, 7, 9\}$ is 5.
 - d. 7 is not the median of the dataset.
 - e. 9 is the maximum value in the dataset, not the median.